

TANJIL HASAN

Rajshahi, Bangladesh | +880 1774 685310 | taanjilhasan@gmail.com | linkedin.com/in/tanjilhasanhimel | tanjilhasanhimel.me

PROFESSIONAL SUMMARY

Final-year Computer Science & Engineering student at Varendra University with a strong record of architecting and directing the build of real, deployed products across web, mobile, and embedded systems using AI-assisted development workflows — alongside two years of live radio broadcasting and hands-on social media management experience. Led the system design of a full-stack digital library platform, architected a multi-tenant print-automation system with real hardware integration, and built a self-hosted 24/7 internet radio platform from scratch. Comfortable moving between deep technical architecture and audience-facing communication work.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript/JavaScript, Dart, Kotlin

Frameworks & Libraries: Next.js, Django, Django REST Framework, Flutter, Jetpack Compose, Tailwind CSS, Room, Hilt, ExoPlayer

Databases & Backend: Supabase (Postgres, Auth, Realtime), MongoDB (Atlas), SQL, SQLite (Room)

Tools & Platforms: Git/GitHub, Vercel, Render, VS Code, MS Office Suite

Core Concepts: System Architecture & App Flow Planning, Multi-Tenancy & Row-Level Security, RESTful API Design, Sensor Fusion & PID Control, AI-Assisted Development

MEDIA & COMMUNICATION SKILLS

Social Media Management: Facebook, Instagram, YouTube, TikTok — content planning, posting, audience engagement

Broadcasting: Live radio hosting, show planning, on-air production coordination

Content & Presentation: Corporate presentation skills, video editing fundamentals, digital marketing fundamentals

FLAGSHIP PROJECTS

Digital Library Ecosystem — Team Lead & System Architect

Nov 2025 – May 2026

Final Year Project, Varendra University

- Architected the system design for a full-stack, multi-tenant digital library platform (supporting multiple departments with isolated data) and directed its AI-assisted development from concept through deployment, leading a project team
- Specified and directed development of a cross-platform Flutter app enabling students to search the catalog, check real-time availability, and borrow books via dynamic QR codes
- Directed the backend and admin dashboard build in Python, Django, and Django REST Framework for real-time inventory and fine management, deployed on Render with MongoDB Atlas
- Designed the logic behind Smart Availability Logic, estimating return dates for checked-out books using historical borrowing data

PrintDAO — Multi-Tenant Print Automation Platform

2026 – Present

Personal Project — System Architect

Live: printdao.vercel.app

- Architected a multi-tenant print-ordering platform for university photocopy shops, directing AI-assisted development of a QR-scan-to-print flow (scan → upload → price → pay → track → collect) requiring no customer account
- Designed a non-FIFO print queue driven by payment-confirmation status rather than arrival order, and a Windows OS-level printer-routing trick that forces lossless black-and-white output at zero added processing cost
- Specified a strict Row-Level-Security architecture for multi-tenancy, designed to prevent malicious users from spoofing page counts, flooding the database, or exhausting storage quotas
- Directed the build of a Python/Tkinter desktop bridge that listens to Supabase Realtime and drives physical printers, with human-in-the-loop approval states for real-money transactions

T Double H FM — Autonomous Internet Radio Station

2026

Personal Project — System Architect & Product Manager

Live: tdoublehfm.vercel.app

- Designed and directed the AI-assisted build of a serverless scheduling engine running a 24/7 autonomous radio station, handling time-zone calculations and day-based programming without exposing raw stream URLs
- Solved browser security constraints (CORS, CSP) for secure playback, and designed a live-override system that instantly pauses the automated schedule the moment I go on-air
- Turned a personal laptop into a live-streaming studio node using Icecast and Cloudflare Tunnels, with a jingle-insertion engine to prevent dead air
- Combined CS training with two years of live radio hosting experience to design the product end-to-end, directing an AI coding agent as the implementation partner

Audia Player — Personal Project

Feb 2026 – Aug 2026

- Directed the AI-assisted design and build of a lightweight, privacy-focused Android music player using Kotlin and Jetpack Compose, integrating ExoPlayer for playback
- Specified an offline-first local database architecture using Room and Hilt for dependency injection, with zero analytics or third-party tracking
- Designed the logic for a “Smart Audio Filter” that automatically detects and excludes junk files (e.g., short WhatsApp voice notes) from the media library
- Specified an “Auto-Pilot” automation feature for scheduled background playback, compatible with Android’s Deep Sleep restrictions

ESP32 Custom Flight Controller Drone — Team Hardware Project

7th Semester

Built from scratch — no off-the-shelf flight controller

- Conceived and directed the AI-assisted development of a custom flight controller from scratch on an ESP32 (instead of off-the-shelf controllers), specifying sensor fusion, motor control, and wireless communication requirements
- Directed implementation of a Madgwick filter and custom PID controller to fuse noisy MPU6050 sensor data into stable flight control, then tuned parameters through hands-on physical flight testing
- Personally calibrated ESC PWM signals and balanced the drone's center of gravity through iterative physical testing for reliable takeoff and hover

ADDITIONAL PROJECTS

Quark Ledger V3 — AI Science News Broadsheet

2026

Early-stage (built over a few days) — refine and validate further before featuring in applications

Live: quarkledger.vercel.app

- Built a small, autonomous news aggregator for physics, robotics, biotech, and space, using AI to synthesize science coverage into formal Standard Bengali
- Used a cron-based round-robin ingestion pattern (Postgres FOR UPDATE SKIP LOCKED) to stay within Vercel's free-tier execution limits while pulling from 40+ RSS sources
- Directed an AI coding agent to build and iterate the system over a few days; article categorization settled into roughly 26 categories after a few rounds of narrowing

The Human Tape — Custom Media Link Platform

Personal Project

- Built a custom media-link hub (an alternative to Linktree) for indie media/YouTube content, with Row-Level-Security-based admin access control
- Directed AI-assisted development of the Next.js/Supabase stack, including a hidden admin route and automated login-alert emails via the Resend API

Noor Travels — Travel Agency CRM

Concept/prototype — not a live product with real users

- Designed a multi-tenant SaaS CRM concept for Hajj/Umrah travel agencies, covering bookings, visa deadlines, and role-based staff access
- Directed AI-assisted development using Next.js, Prisma, and PostgreSQL as a demonstration of the concept, not a deployed production system

Smaller builds

Import Export Hub — MERN logistics platform for trade documentation and shipment tracking

GamerHat — React/Redux e-commerce site with Node.js and Stripe — live: gamerhat.netlify.app

Digital Life Lessons — Node/Express/MongoDB edtech platform for digital literacy content

RadioBang — Broadcast website for live radio streaming (Icecast/Butt-based)

Other utility builds — Happy Birthday (interactive greeting generator), BoRoCare (early-stage care-services concept, frontend only), LotsHero Apps (small HTML/CSS/JS utility collection), CS Ticket (frontend support-ticket UI), Emergency Services (numbers directory)

EDUCATION

B.Sc. in Computer Science & Engineering

Sep 2022 – Expected 2026

Varendra University, Rajshahi — Final Semester (8th Semester)

EXPERIENCE

Radio Jockey & Program Presenter

Jan 2025 – Present

Radio Padma, Rajshahi, Bangladesh — promoted from Intern (Feb 2024 – Jan 2025)

- Host live radio programs and manage on-air content, developing strong real-time communication and composure under pressure
- Coordinate with production teams and adapt quickly to last-minute changes during live broadcasts

Media Manager

~1.5 months, 2025

Delupi Film

- Managed day-to-day content and posting across Facebook, Instagram, YouTube, and TikTok for film promotion campaigns
- Handled the majority of hands-on social media execution, with final creative and strategic decisions made jointly with the production team

LEADERSHIP & ACTIVITIES

- Youth MP, Youth Parliament (Oct 2023 – Jan 2026) — Participated in policy discussions and civic engagement, strengthening public communication and teamwork

CERTIFICATIONS

Technical & AI

- CS50: Introduction to Computer Science — Harvard University (edX)
- Full Stack Development with MERN — Grameenphone Academy
- Artificial Intelligence & Machine Learning Fundamentals — Grameenphone Academy
- Human-in-the-Loop Data Refinement & Annotation — Certificate of Appreciation
- Robotics: Aerial Robotics — Coursera
- IT/Hi-Tech Park Development — Bangladesh Hi-Tech Park Authority (ICT Division), conducted by Gurukul Institute of IT

Business, Leadership & Communication

- Aspire Leaders Program — Aspire Institute
- Corporate Presentation Skills — Grameenphone Academy
- Build Your Own Startup — Robi Axiata PLC
- Write Professional Emails in English — Coursera
- Bystander Youth Leadership Training — Rupantor

Creative & Media

- Professional Video Editing — Futurenation
- UI/UX Design — Futurenation
- Introduction to Digital Marketing — Bohubrihi
- Fundamentals of Digital Marketing — Google
- Moonsoon Revolution Film Production Workshop — Ministry of Cultural Affairs & Bangladesh Shilpakala Academy